

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 10%

QUESTIONS: 2

Duration : 45 Minutes
Total Marks:20

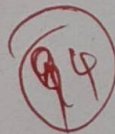
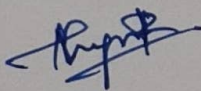
Annexure A

Descriptive Written Test

Code of Conduct:

I R. Akshaya (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Annexure A

Descriptive Written Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.
2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem. Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.
3. A smart traffic management system is being designed to control traffic signals across a city based on vehicle density, pedestrian movement, and weather conditions. The system must respond to real-time sensor inputs while also learning traffic patterns over time to reduce congestion. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for minimizing congestion and travel time.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

ARTIFICIAL INTELLIGENCE

CSA-1710

COI-ATS.

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① Intelligent Agent in Smart Home Automation System:

* Intelligent Agent is a system that observes the environment through sensors and performs the actions through actuators.

* In smart home it includes the sensors such as motion detection, temperature changes, and time of day while also learning user preferences over time.

Types of Intelligent Agents:

The types of intelligent agent includes

1. Simple Reflex Agent
2. Model Based Agent
3. Goal Based Agent
4. Utility Based Agent.

1. Simple Reflex Agent:

⇒ It acts only on the current input using predefined rules and not with the previous Events

⇒ Eg: Turning on the lights when motion is detected, switches off lights when motion is not detected.

2. Model based Agent:

⇒ A model based agent maintains an internal model of the Environment by remembering the previous Events.

⇒ Eg: Adjust room temperature based on previous temperature readings.

3. Goal based Agent:

⇒ A goal based agent chooses actions that helps to achieve a desired goal.

⇒ Eg: Ensures all doors are locked before bed time.

4. Utility-Based Agent :-

⇒ A Utility-Based Agent selects the action that provides the highest overall benefit by considering multiple factors.

⇒ Eg: Balances user comfort with electricity consumption. Gives higher priority to security during night-time.

Analysis of Agent types :

Agent Types	Functions in Smart Home
Simple Reflex Agent	Performs immediate actions like switching lights on/off.
Model-Based Agent.	Uses previous information to improve decisions.
Goal-Based Agent	Plans actions to achieve goal as maintaining comfort and security.
Utility-Based Agent	Selects the best action by balancing comfort, energy efficiency and safety.

Most Effective Agent:

⇒ A hybrid Approach combining Model-Based, Goal-based, Utility based Agent is the best solution.

Justification:

- * Comfort: Learns user preferences & maintains ideal temperature and lighting.
- * Energy Efficiency: Reduces unnecessary power consumption.
- * Security: Detects suspicious activities and responds appropriately.

Thus, a hybrid intelligent agent provides the highest efficiency, adaptability and user satisfaction.

2. Smart Traffic Management System Using Intelligent Agents.

⇒ A smart traffic management system controls traffic signals based on vehicle density, pedestrian movement, and weather conditions.

⇒ It receives data from cameras, sensors, and traffic lights, while learning traffic patterns to reduce congestion and improve road safety.

1. Simple Reflex Agent:

A simple Reflex Agent makes decision based only on the current input using predefined rules.

⇒ Eg: 1. Turns the traffic light green when vehicle density is high.

2. Gives pedestrian crossing when pedestrian button is pressed.

3. Switches to red if an emergency vehicle is detected.

2. Model-based Agent:

A model-based Agent maintains an internal model of the Environment by storing previous informations.

- ⇒ Eg:
1. Remembers traffic flow at different times
 2. Predicts congestion using previous traffic data.

3. Goal-based Agent:

A Goal-based agent selects actions to achieve a specific goal.

- ⇒ Eg:
1. Minimizes traffic congestion.
 2. Reduces waiting time at signals.

4. Utility-based Agent:

A Utility based Agent chooses the action that provides the highest overall benefit by evaluating different outcomes.

- ⇒ Eg:
1. Balances traffic flow, safety, and fuel consumption
 2. Adjusts traffic lights during bad weather to improve safety.

Analysis of Agent Types

Agent Type	Function in Traffic Management
Simple Reflex Agent	Responds immediately to sensor inputs.
Model-Based Agent	Uses previous traffic information for better decisions.
Goal-based Agents	Plans actions to reduce congestion and improve safety.
Utility-Based Agent	Chooses the best action by balancing traffic flow, travel time, and safety.

Most Effective Agent:

A hybrid approach combining Model-based, Goal based, and Utility-based Approach is the most Effective.

Justification:-

- * Model-based Agent remembers traffic patterns and predicts congestion.
- * Goal-based Agent focuses on reducing travel time and improving safety.
- * Utility-based Agent selects best signal timings by balancing traffic flow.

Conclusion :

A hybrid Intelligent Agent is the best choice for Smart traffic Management System because it can learn from past traffic data, achieve traffic management goals and make optimal decisions in real time.

Working :-

- * The robot starts from the Entrance.
- * It scans nearby paths using its sensors.
- * It record to an unexplored location.
- * It records visited path and builds an internal map.
- * If it reaches a dead End, it backtracks and explores another route.
- * The process continues until the Exist is found.

Map Building:-

- * Stores visited locations
- * Marks blocked paths and dead Ends.
- * Updates connections between Explored areas
- * Gradually creates a complete map of cave

Risks of irreversible Actions:-

- * Entering a one way tunnel.
- * Falling into a pits or unstable areas
- * Closing passages permanently.
- * Running out of battery before finding the Exist.

Conclusion :

⇒ Online Search Enables that robot to Explore unknown Environments while learning from Experience.

⇒ However, irreversible actions may trap the robot or prevent it from reaching the Exit.

2. Solve the 6-Queens problem: place 6 queens on a 6x6 chessboard such that no two queens attack each other. Formulate the problem components (state space, initial state, successor function, goal test) and Explain how incremental formulation reduces the space search compared to complete state formulation.

Ans:-

One Valid Solution:

Row	Column
1	2
2	4
3	6

Row	Column
4	1
5	3
6	5

No two queens attack each other horizontally, vertically, or diagonally.

Problem Formulation:

State Space:

All possible arrangements of queens on a 6×6 chessboard.

Initial State:

Empty chessboard (no queens placed).

Success Function:

Place one queen in the next row in a safe column where it is not attacked by any previously placed queen.

Goal Test:

8 queens are placed safely on the board without any conflicts.

Incremental Formulation:

In incremental formulation:

1. Queens are placed one row at a time.
2. Unsafe positions are rejected immediately.
3. Only valid partial solutions are expanded.

Advantages Over Complete-State Formulation:

- * Reduces search space significantly.
- * Avoids checking all $6^6 = 46,656$ possible board arrangements.
- * ~~Faster~~ and more memory-efficient.

3. A hospital wants to build an AI agent to triage incoming patients based on symptom severity. Define an appropriate performance measured for this agent. and Explain the concept of rationality in this context - under what conditions would the agent's action be considered rational?

Ans :- AI Agent for Hospital Patient Triage:

Performance Measure:

The AI agent should:

- * Correctly identify emergency patients.
- * Minimize waiting time.
- * Reduces incorrect diagnosis.
- * Prioritize patients based on severity.
- * Improve patient safety and treatment efficiently.

Rationality :

A rational agent selects action that maximizes the Expected performance based on the available information.

The agent is considered rational when it:

- ⇒ Uses patient symptoms and medical history.
- ⇒ Correctly estimate the severity level.
- ⇒ Prioritize critical patients immediately.
- ⇒ Follows medical guidelines.
- ⇒ Updates decisions when new information

Example: If a patient has severe chest pain, low blood pressure, and difficulty breathing, a rational AI agent immediately assigns high priority for emergency treatment instead of making the patient wait.

Conclusion: A rational hospital triage agent makes decision that maximize patient safety, improve treatment quality and efficiently utilize hospital resources.